A programmer's walkthrough of inter-universal Teichmüller theory: running the multiradial algorithm on explicit initial Θ-data

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Abstract

We describe a program, with an interactive web page, that runs on explicit numbers those operations of inter-universal Teichmüller theory (IUT) that can be written as procedures. Its input is one of seven explicit initial Θ-data from an earlier catalogue. The program computes the Tate parameters and the local data p-adically, arranges the labels and processions, and keeps the two Hodge theaters of a Θ-link as copies of the local fields separated by type, so that only a "cable"—the unit-group data and one unlabelled value-group object—passes between them, after an analogy that the source itself draws. It then computes log-shells, Kummer read-outs, the containers of the multiradial representation and the possible images of the theta values, and evaluates the log-volume of the holomorphic hull of their union, as an exact combination of logarithms, for three readings of the label indeterminacy and two models of the indeterminacy (Ind2): multiplication by units, and all $\mathbb{Z}_p$-linear automorphisms of the log-shells of the factors, as in the Lean formalization lana-agents/iut. For the second model the hull is given by a closed form from the author's earlier note; it is exact where p is odd and $e \leq p-2$, and bracketed elsewhere. Finally it places side by side the two quantities that Step (xi) of the proof of Corollary 3.12 of IUT III compares. In the reading of IUT IV the two models give the two forms of the comparison that lana-agents/iut calls the Scholze–Stix inequality and the variant of Corollary 3.12, and the program computes their difference for each datum; version 1.1.0 adds the second model. Infinite objects are computed in finite windows, and the program prints how its output stabilizes as the windows grow. The properties IPL, SHE and APT of Remark 3.11.1 of IUT III appear as the shape of the code and as records of its execution; the inference (xi-f) is displayed, not evaluated. The purpose is expository. No conclusion is drawn about Corollary 3.12, Theorem 1.10 of IUT IV, or the abc conjecture. Turning the objects and processes of IUT into data types, with IPL and APT proved for a Lean model, has been done earlier and on a broader scale in the toolkit motizukipdf; the formulas for the containers and the distinction between the readings are taken from the Lean formalization lana-agents/iut.

1 Introduction

The four papers on inter-universal Teichmüller theory [IUT1, IUT2, IUT3, IUT4] are long, and much of their length goes into showing that each operation can be carried out on group-theoretic data alone, without the ring structures of the objects it acts on, and that the operations are compatible with one another. The operations themselves can be described as procedures acting on an initial Θ-datum: build labels and Hodge theaters, compute theta values, pass data through the Θ-link and the log-link, place the theta values into containers up to certain

indeterminacies, take a hull, and compare two log-volumes. The source itself invites this view. Remark 3.11.1(iv) of [IUT3] compares the theta functions and theta values used by the multiradial algorithm to the internal structure of a CPU or an operating system, which matters for how the algorithm runs but not for an "end user" who only feeds it input data and reads its output, and it compares the prime-strip of the Θ-link to a cable connecting separate computers. Mochizuki's report on the formalization of IUT [Formal, §1] describes formalization as a tool for communicating the logical structure of the theory, rather than mainly as a tool for verification, and its overview [Formal, §3] states the main point in terms that a programmer can take literally: the multiplicative monoid structures and the abstract groups are common to the two sides of the Θ-link, while the ring structures are not, so that the former can serve as input data for a common container of the two ring structures.

This note describes a program that runs those procedures on explicit data, for a reader who knows programming better than anabelian geometry. It does not reproduce the proofs of the source. It follows three rules (Section 2): each Hodge theater holds its own copy of the relevant fields, and the type system refuses arithmetic between copies; every infinite object is computed in a finite window whose growth is displayed; and the readings of the source that differ are selected at one place. With these rules, the properties of the multiradial algorithm that Remark 3.11.1 of [IUT3] singles out—the input prime-strip link (IPL), simultaneous holomorphic expressibility (SHE) and algorithmic parallel transport (APT)—become properties of the code that one can watch.

The data are the seven explicit initial Θ-data of the catalogue [Cat], with field of moduli $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(i)$. The last step of the program is the comparison in Step (xi) of the proof of Corollary 3.12 of [IUT3]: it computes both sides and reports whether the computed numbers satisfy the inequality. In the reading in which the theta value stays at its label, as in Step (v) of the proof of Theorem 1.10 of [IUT4], the comparison has the form $(s-1)\,|\log(\underline{\underline{q}})| \le V$ of the inequality that the Lean formalization [LANA-iut] calls the Scholze–Stix inequality [SS], where V is the log-volume of the container (Section 3.7); the same reduction is the toy form in [Formal, §3], by which a height that is multiplied by N but unchanged up to a mild error is bounded. If (Ind2) acts through all $\mathbb{Z}_p$-linear automorphisms of the log-shells of the factors, as in [LANA-iut], the comparison becomes $(s-1)\,|\log(\underline{\underline{q}})| \le V + \Delta(D)$, the form of the variant of Corollary 3.12 there, with a gap $\Delta(D) > 0$ that the program computes by the closed form of [Gap]. That the inequality holds on data of small height is expected whatever the status of Corollary 3.12 (for D_2 it was checked in [Cat, §5]), and we draw no conclusion from it.

What is not new. The objects and processes of IUT I–IV have been turned into data types, with a registry of processes, a web application and a Lean port, in the toolkit motizukipdf [Richards]. Its data model lets a Θ-link share the unit groups but never the ring structures, and records (Ind1) as permutations of tensor factors; its Lean port, following [Formal], proves IPL and APT for its model and derives a bound $|\log(\underline{\underline{q}})| \le H/(N-1)$, with H its estimate of the hull term and $N = \ell(\ell+1)/12$, from the comparison in the common container, which it leaves as a hypothesis. Its C code also evaluates a numerical form of that comparison, following [EssLgc, §3.10]: the Θ-pilot is widened by the indeterminacies to an interval whose upper end is given by a leading-term estimate of the hull, and the program reports whether the interval contains $-|\log(\underline{\underline{q}})|$. The formula we use for the log-volume of the hull of a tensor packet of log-shells, the two readings of the label indeterminacy, and the observation that under the literal reading the hull is governed by a minimum rather than an average are in the Lean repository [LANA-iut], which also proves, in its models, that Step (xi) is equivalent to the Scholze–Stix inequality. That (Ind1) permutes tensor factors is also noted in [DH1, §4] and [DH, §6.2, footnote 7]. The data and their local invariants are those of [Cat]. Section 8 gives details.

What this note adds. A program in which these pieces run together on verified explicit data whose field of moduli is not $\mathbb{Q}$; the value of the hull term for each datum and reading, computed by enumerating the tuples of places where they are few and otherwise by a closed form that is checked against the enumeration; finite windows and the stabilization of the output; IPL, SHE and APT as records of the execution of one program on numbers; and two visual presentations of the same computation. Version 1.1.0 adds a second model of (Ind2), the group of all $\mathbb{Z}_p$-linear automorphisms of the log-shells of the factors used in [LANA-iut], whose hull is computed by the closed form of [Gap], and with it the gap between the two forms of the comparison for each datum and reading (Table 4).

Contents. Section 2 states the design rules. Section 3 follows the steps from the initial Θ-datum to Step (xi). Section 4 lists the windows and Section 5 the results on the seven data. Section 6 describes the web page, Section 7 separates what is computed from what is given or chosen, and Section 8 discusses related work. Appendix A explains how to reproduce the results and Appendix B maps the code to the source.

2 Design

2.1 Theaters as type-separated copies

A Hodge theater is an object of class `Theater`. It owns its own copy of the local fields K_v. An element carries a reference to the theater it belongs to, and the product of two elements of different theaters raises a type error ("the ring structures are different"). The Θ-link ([IUT1, Cor. 3.7], [IUT3, Def. 3.8]) is the function `theta_link`, which does not map values. It builds a `Cable`, the data of an $\mathcal{F}^{\vdash\blacktriangleright\times\boldsymbol{\mu}}$-prime-strip, and attaches it to both theaters:

- the unit part ($\mathcal{F}^{\vdash\times\boldsymbol{\mu}}$): for each place, the group $\mathcal{O}_v^\times/\boldsymbol{\mu}$ as an abstract object, determined by the residue characteristic, the ramification index and the residue degree;
- the value-group part ($\mathcal{F}^{\vdash\blacktriangleright}$): one unlabelled object λ, realized in the domain theater by the Θ-pilot and in the codomain theater by the q-pilot.

The two realizations of λ have degrees $s\,|\log(\underline{\underline{q}})|$ and $|\log(\underline{\underline{q}})|$ measured in the two theaters, where $s = \ell(\ell+1)/12$ (Section 3.3). Nothing that depends on the ring structure of one theater is available in the other. This is the main point of [Formal, §3] recalled in the introduction, and the data model of [Richards] treats the Θ-link in the same way. The name of the class follows the analogy of [IUT3, Rem. 3.11.1(iv)].

2.2 Finite windows

Every infinite object is computed in a finite window: p-adic numbers to a given number of digits, the Kummer theory of a local field at levels $n \mid p-1$, the factorization of the global data within a given trial bound and time, the logarithm series to a given number of terms, and the sublattices admitted by (Ind3) up to a given depth. The program reports its output for growing windows, and the reader extrapolates from the part that has stopped changing (Section 4).

2.3 One switch for the readings

The only input that encodes a choice of reading of the source is the set of factors of the capsule $\mathbb{S}^{\pm}_{j+1} = \{0,\dots,j\}$ on which the theta value of label j may be placed under (Ind1) before the hull is taken:

$$P_{\mathrm{V}}(j) = \{j\}, \qquad P_{\mathrm{L}}(j) = \{1,\dots,j\}, \qquad P_{\mathrm{L}^0}(j) = \{0,\dots,j\}.$$

These are the readings (V), (L) and (L^0) of [Cat]. Reading (V) keeps the theta value at its label, as in Step (v) of the proof of Theorem 1.10 of [IUT4]; it is the reading of IUT IV in [LANA-iut], which also records that taking the hull for each image under (Ind1) separately gives this value. Reading (L^0) admits every position of the capsule; it is the literal reading of [LANA-iut] (file `Iut/Concrete/Literal.lean`). Reading (L) admits the positions $\{1, \dots, j\}$, that is, the label permutations fixing 0; this set is the $\mathbb{F}_\ell^*$-capsule $\mathbb{S}_j^*$, which [IUT1, Prop. 6.9(iii)] includes in $\mathbb{S}_{j+1}^{\pm}$. We do not claim that any of these is the reading the source intends.

The group by which the indeterminacy (Ind2) acts is a second parameter, with two values (Section 3.6). It is a choice of model rather than a reading of the label indeterminacy, and the program reports both values side by side.

2.4 IPL, SHE and APT as the shape of the code

Remark 3.11.1(iii)–(v) of [IUT3] describes the output of the algorithm of Theorem 3.11 by three properties, which we paraphrase. *IPL*: the output data are linked to the input prime-strip by the full poly-isomorphisms of $\mathcal{F}^{\Vdash\blacktriangleright\times\mu}$-prime-strips induced by the operations of the algorithm. *SHE*: the construction of the output, and the output itself, are expressed in a vocabulary that is valid simultaneously for the arithmetic holomorphic structure that gives rise to the Θ-pilot in the domain of the Θ-link and for the one that gives rise to the input prime-strip in its codomain. *APT*: this "parallel transport" is not the transport of a set-theoretic region by a set-theoretic function, but a construction algorithm that is valid simultaneously for both arithmetic holomorphic structures. In the program:

- *APT* is the fact that the same function `IUTAlgorithm.run(cable, etale, ring)` is executed with the ring of the domain theater and with the ring of the codomain theater; no value is carried from one theater to the other by a function.
- *SHE* is recorded by logging which theater's ring the run uses: in the run with the codomain ring, the domain ring is never used. Moreover, the algorithm does not read the value of its input, so replacing the input q-pilot by a q^2-pilot, as [IUT3, Rem. 3.11.1(ii)] allows, leaves the output unchanged. We take this as the program's counterpart of Step (xi-e) of [IUT3, Cor. 3.12, proof], where SHE is read as expressibility by operations that remain valid when the pilot log-volume of the input is fixed; the source does not phrase it in these terms.
- *IPL* is the fact that the output carries a reference to the very object λ of the cable.

These records hold by construction, since the ring is used only to realize the containers as lattices. They show the structure; they prove nothing.

3 From the initial Θ-datum to Step (xi)

3.1 Initial Θ-data

A datum is given by $D \in \{5, -1\}$, an element $\pi = x + y\sqrt{D}$ of $F_{\mathrm{mod}} = \mathbb{Q}(\sqrt{D})$ of prime norm p, integers m, k and a prime ℓ. The curve is $E\colon y^2 = x(x-1)(x-a)$ with $a = \pi^m \bar{\pi}^k$, and the fields F and K of [IUT1, Def. 3.1] are those of [Cat]; the set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ is described below. The program computes:

- the odd places w of F_{mod} at which $\mathrm{ord}_w(a) \neq 0$ or $\mathrm{ord}_w(1-a) > 0$, which are those where $j = 2^8(a^2 - a + 1)^3/(a^2(a-1)^2)$ has $\mathrm{ord}_w(j) < 0$, and $\mathrm{ord}_w(q) = -\mathrm{ord}_w(j)$ for the Tate parameter q; finding them requires factoring the norms, which is done in a window;
- at the places above p of multiplicative reduction, the Tate parameter itself, p-adically, from $t = 1/j$ and the series $q = t + 744t^2 + 750420t^3 + 872769632t^4 + \cdots$ obtained by

inverting $t = \Delta(q)/E_4(q)^3$;

- at a bad place w, the ramification index of K over $\mathbb{Q}_p$, taken to be $e(F_{\mathrm{mod},w}/\mathbb{Q}_p)\cdot 15\ell/\gcd(15\ell, \mathrm{ord}_w\, q)$. At the places above p, where $F_{\mathrm{mod},w} = \mathbb{Q}_p$, this is the formula used in the proofs of [Cat, App. A], and it reproduces Table 1 there; at the other bad places the program applies it as an assumption. At good places outside $2\cdot 3\cdot 5\cdot \ell$, where K/F_{mod} is unramified, the index is $e(F_{\mathrm{mod},w}/\mathbb{Q}_p)$;
- the conditions of [IUT1, Def. 3.1] that can be decided from these numbers: $\ell \geq 5$ is prime; $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ is nonempty and its residue characteristics are odd and different from ℓ; $\ell \nmid \mathrm{ord}_w(q)$; the reduction at $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ is multiplicative; and, as a consistency check, the valuation of $\underline{\underline{q}}_w = q_w^{1/2\ell}$ [IUT1, Ex. 3.2(iv)] is integral in K_w.

As $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ the program takes the places of multiplicative reduction found in the window outside the primes $2, 3, 5, \ell$, with $\ell \nmid \mathrm{ord}_w\, q$; it leaves out the places above 3 and 5 because the ramification of K there depends on the construction of F. This is a subset of the set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ of [Cat], which consists of all places of multiplicative reduction outside 2ℓ; for five of the seven data the two differ by places above 3 or 5, and $|\log(\underline{\underline{q}})|$ is lower by at most 0.28. Definition 3.1 allows a subset (it notes that places outside $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ may have bad reduction), and each window gives a valid choice. The local data above p reproduce Table 1 of [Cat] for all seven data.

3.2 Labels and processions

The program builds $\mathbb{F}_\ell$, $|\mathbb{F}_\ell| = \{0, \ldots, \ell^*\}$ and $\mathbb{F}_\ell^* = \{1, \ldots, \ell^*\}$ with $\ell^* = (\ell-1)/2$, lets $\mathbb{F}_\ell^{\gtrdot\pm}$ act additively and $\mathbb{F}_\ell^*$ multiplicatively, and forms the procession $\mathbb{S}_1^\pm \subseteq \cdots \subseteq \mathbb{S}_{\ell^\pm}^\pm$, $\mathbb{S}_t^\pm = \{0, \ldots, t-1\}$, $\ell^\pm = \ell^* + 1$ ([IUT1, Def. 4.10, Props. 4.11, 6.9]). By [IUT1, Prop. 6.9(i)], each index of the n-capsule has exactly n possibilities, so that the processions reduce the $(\ell^\pm)^{\ell^\pm}$ label indeterminacies of [IUT1, Prop. 6.8(ii)] to $\ell^\pm!$; the page shows both counts for each datum.

3.3 Pilots and the Θ-link

For a bad place w put $r_w = \mathrm{ord}_p(\underline{\underline{q}}_w) = \mathrm{ord}_w(q)/(2\ell\, e(F_{\mathrm{mod},w}/\mathbb{Q}_p))$ and give w the weight $c_w = [F_{\mathrm{mod},w} : \mathbb{Q}_p]/[F_{\mathrm{mod}} : \mathbb{Q}]$. The global arithmetic degree of an object of the realified global Frobenioid is its log-volume ([IUT3, Prop. 3.9(iii)], as recalled in Step (xi-d)), and the program computes

$$|\log(\underline{\underline{q}})| = \sum_{w\in\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}} c_w\, r_w \log p_w, \qquad \frac{1}{\ell^*}\sum_{j=1}^{\ell^*}\sum_{w\in\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}} c_w\, j^2 r_w \log p_w = s\,|\log(\underline{\underline{q}})|, \quad s = \frac{\ell(\ell+1)}{12},$$

for the q-pilot and the Θ-pilot $\{\underline{\underline{q}}^{j^2}\}_j$. These are the two realizations of the object λ of the cable, and their ratio s is the dilation of the Θ-link. All log-volumes are kept exactly as rational combinations of logarithms of primes and of $\log\pi$; they are evaluated numerically only to display them and to decide the outcomes (Section 3.7).

3.4 The log-link and log-shells

The log-link ([IUT3, Def. 1.1]) sends units to the additive group. The program computes the p-adic logarithm of the unit part u of the Tate parameter at the places above p, as $\log(u^{p-1})/(p-1)$, and the log-shell $\mathcal{I}_v = (2p)^{-1}\log(\mathcal{O}_v^\times)$; for odd p this is the log-shell of [IUT3, §1], and it is the normalization used in the proof of [IUT4, Thm. 1.10, Step (v)]. Its radius is

$$\rho_v = -\min\{\mathrm{ord}_p\, z : 0 \neq z \in \mathcal{I}_v\} = 1 + \mathrm{ord}_p(2) - \frac{m^*}{e}, \qquad m^* = \min_{s\geq 0}(p^s - s\,e),$$

provided the minimum defining m^* is attained at a single s: then, since $\mathcal{O}_v^\times = \boldsymbol{\mu} \times (1 + \varpi\mathcal{O}_v)$ and the term x^k/k of $\log(1+x)$ has ϖ-adic valuation $k\,\mathrm{ord}_\varpi(x) - e\,\mathrm{ord}_p(k)$, the term of least valuation decides. When the minimum is attained twice, cancellation can raise the valuation (for instance when $e = p-1$ and K_v contains the p-th roots of unity); this happens at no place of our data. For $e \le p-2$ this gives $\rho_v = 1 - 1/e$, the value in [LANA-iut]. The program checks the formula by summing the series for $\log(1+2\varpi)$ in a model $\varpi^e = p$ of the totally ramified part. For example, at the places above 11 of the datum D_2, where $e = 105 > 11 - 2$, the series has valuation -94 and $\rho = 199/105$. The container is made from units only, which is why it is the same in both theaters.

3.5 Kummer read-out

At a level $n \mid p-1$, the Kummer class of an element $\alpha = p^a u$ of $\mathbb{Q}_p^\times$ is determined by two values of the tame Hilbert symbol, and its inertia coordinate is $-a \bmod n$. The program reads $\mathrm{ord}(q^{j^2}) \bmod n$ from the Kummer classes of the elements q^{j^2} of $\mathbb{Q}_p^\times$, whose 2ℓ-th roots in K_v are the theta values, and combines the levels by the Chinese remainder theorem; the read-out stops changing once the modulus exceeds the value. For D_2 the values $12, 48, 108$ are fixed once the level 11 is included (modulus 110). For D_{211}, where $p-1 = 210$, the levels available in $\mathbb{Q}_{211}$ give the values only modulo 210, and residue field extensions would be needed. The class is computed from the valuation, so the read-out is a round trip that illustrates how the group-theoretic class determines the valuation; it is not an independent check.

3.6 The multiradial algorithm

By [IUT3, Thm. 3.11(i)(a)], the container of the capsule of label $j \in \mathbb{F}_\ell^*$ is the tensor product of the log-shells over $\mathbb{S}^{\pm}_{j+1} = \{0, \dots, j\}$. As in [IUT4, Thm. 1.10, proof, Step (iv)], at a prime p it splits into components indexed by tuples $T = (v_0, \dots, v_j)$ of places above p, and the components are averaged with the weights $c_T = \prod_t c_{v_t}$. In a component, the holomorphic hull of the tensor product of the log-shells has log-volume $\sum_t \rho_{v_t} \log p$: this is a theorem of [LANA-iut] (`componentVol_leastHull_tensorLogShell_eq`, file `Iut/WallEquiv/VcI.lean`), and Step (v) of [IUT4, Thm. 1.10, proof] bounds the same component from above by Proposition 1.4(iii) there. The prime p therefore contributes $(j+1)\bar\rho_p \log p$ to the container, with $\bar\rho_p = \sum_{v\mid p} c_v \rho_v$. At the archimedean places we use the container $\pi^{j+1} \cdot B_I$ of [IUT4, Thm. 1.10, proof, Step (vii)], of log-volume $(j+1)\log\pi$; [LANA-iut] uses the larger unit ball of [IUT3, Prop. 3.2(ii)], which would only increase the values of V below.

The possible images of the Θ-pilot of label j in the component T are modelled as the regions $\underline{\underline{q}}_{v_t}^{j^2} \cdot$ (tensor product of the log-shells) for $t \in P_R(j)$, the positions allowed by (Ind1) in the reading R, moved by the indeterminacy (Ind2). The program has two models of (Ind2). In the first, (Ind2) acts by multiplication by units, which does not move these lattices, so that the hull of the union is the one for the identity alone. In the second, (Ind2) acts through all $\mathbb{Z}_p$-linear automorphisms of the log-shells of the factors, the group of [LANA-iut] (file `Iut/Concrete/ThetaRegion.lean`), which is stated there to contain the image of the group Ism of the source and to be in general larger, a deliberate enlargement. The hull for the group of the source therefore lies between the two. The indeterminacy (Ind3) admits sublattices, which do not change the hull in either model.

In the first model, multiplication by an element of K_{v_t} of p-adic valuation c moves the hull of the tensor packet to a principal ideal of its maximal order whose exponents are all proportional to c, so the hulls of the possible images in a component are totally ordered; the hull of their union is the largest of them, and its log-volume is the container minus

$$\min_{t \in P_R(j)} j^2\, r_{v_t} \log p.$$

In the second model, put $\kappa_v(\theta) = \max\{k \in \mathbb{Z} : \theta\,\mathcal{I}_v \subseteq p^k \mathcal{I}_v\}$ for $\theta \in K_v^\times$. The images of $\theta \cdot$ (tensor product), with $\theta = \underline{\underline{q}}_{v_t}^{j^2}$ in the factor t, span $p^{\kappa_{v_t}(\theta)}$ times the tensor product, since the automorphisms of the lattice $\mathcal{I}_{v_t}$ move a primitive element of $\theta\,\mathcal{I}_{v_t}$ to every vector of a basis, and the hull depends only on the $\mathbb{Z}_p$-span [Gap, Lemmas 4.1–4.3]. The union over $t \in P_R(j)$ spans $p^{\min_t \kappa}$ times the tensor product, so the hull of the union has log-volume the container minus

$$\min_{t \in P_R(j)} \kappa_{v_t}\big(\underline{\underline{q}}_{v_t}^{j^2}\big) \log p;$$

in reading (V) this is the closed form of [Gap, Thm. 3.2]. One has

$$\lfloor \mathrm{ord}_p\,\theta - c_v \rfloor \;\le\; \kappa_v(\theta) \;\le\; \lfloor \mathrm{ord}_p\,\theta \rfloor, \qquad c_v = \min\big(a_v + b_v,\; \rho_v - \rho_v'\big),$$

where a_v, b_v are the constants of [IUT4, Prop. 1.2(i)] (see [Gap, Thm. 3.3]), and $\rho_v' = 1 + \mathrm{ord}_p(2) - n_0/e$ with $n_0 = \lfloor e/(p-1) \rfloor + 1$: since $\log(1+\mathfrak{m}^n) = \mathfrak{m}^n$ for $n > e/(p-1)$, every element of K_v of valuation at least $-\rho_v'$ lies in $\mathcal{I}_v$, and every element of $\mathcal{I}_v$ has valuation at least $-\rho_v$. At a *small* place, where p is odd and $e \le p-2$, one has $c_v = 0$, and $\kappa_v(\theta) = \lfloor \mathrm{ord}_p\,\theta \rfloor$ exactly. At the other places the program brackets $\kappa_v(\theta)$, whose value depends on the unit part of θ and on K_v [Gap, Ex. 5.2].

The program computes the weighted sums over the tuples by enumeration when there are at most 256 of them, and otherwise by the closed form $\sum_k (f_{(k)} - f_{(k-1)})\big(\sum_{i \ge k} c_{(i)}\big)^{|P_R(j)|}$ over the sorted values $f_{(1)} \le f_{(2)} \le \cdots$ $(f_{(0)} = 0)$ of $f(v) = j^2 r_v$ in the first model, and of each end of the bracket for κ_v in the second: the expected value of the minimum of $|P_R(j)|$ independent choices. Averaging over j gives

$$-|\log(\underline{\underline{\Theta}})| \;\ge\; V - \Sigma_R, \qquad V = \frac{1}{\ell^*}\sum_j \mathrm{container}_j, \quad \Sigma_R = \frac{1}{\ell^*}\sum_j \mathrm{shrinkage}_j,$$

where Σ_R is computed in the first model, or in the second model with the upper ends $\lfloor \mathrm{ord}_p\,\theta \rfloor$, which give the largest shrinkage; the inequality accounts only for the places that are not used (above $2, 3, 5, \ell$, and above any part of the global data not factored in the window), whose contributions in the model are nonnegative. In reading (V) and the first model, $\Sigma_{\mathrm{V}} = s\,|\log(\underline{\underline{q}})|$. The difference of the shrinkages of the two models is the gap

$$\Delta_R(D) = \Sigma_R^{\mathrm{units}} - \Sigma_R^{\mathrm{lattice}} \;\ge\; 0,$$

which the program computes as an interval from the two ends of the bracket; it is exact when every bad place is small. In reading (V) it is the gap $\Delta(D)$ of [Gap] (Section 3.7).

3.7 Integration: Step (xi)

Step (xi) of the proof of Corollary 3.12 of [IUT3] presents "two tautologically equivalent ways" of computing the log-volume of the q-pilot object of the Hodge theater $(1,0)$ (Fig. 3.8 there):

A. the q-pilot of $(1,0)$, measured directly in $(1,0)$: $-|\log(\underline{\underline{q}})|$;

B. the Θ-pilot of $(0,0)$, carried by the multiradial algorithm, followed by the holomorphic hull and the log-volume, to the column of $(1,0)$: the set $\mathbb{R}_{\le -|\log(\underline{\underline{\Theta}})|}$ of possible log-volumes.

In Step (xi-e), SHE is read as expressibility by operations that remain valid when the pilot log-volume of the input prime-strip is subject to the condition of being equal to the fixed value $-|\log(\underline{\underline{q}})|$; by IPL, the input is linked to the possible outputs by isomorphisms of $\mathcal{F}^{\vdash\blacktriangleright}$-prime-strips. Step (xi-f) infers that the construction of $\mathbb{R}_{\le -|\log(\underline{\underline{\Theta}})|}$ is subject to the condition that it also be, possibly up to an "approximation" by the indeterminacies, a construction of the pilot log-volume $-|\log(\underline{\underline{q}})|$ of the input, so that $-|\log(\underline{\underline{q}})| \in \mathbb{R}_{\le -|\log(\underline{\underline{\Theta}})|}$. The program displays

this inference as a rule and does not evaluate it. It computes the two numbers and reports one of three outcomes: the inequality holds with the container of the factored places alone; it holds after adding a lower bound for the contribution of the unfactored part of the global data; or it is undecided in the window. It never reports a failure, because the computed value of $-|\log(\underline{\underline{\Theta}})|$ is only a lower bound.

The comparison is equivalent to $\Sigma_R - |\log(\underline{\underline{q}})| \leq V$. In reading (V) it is $(s-1)\,|\log(\underline{\underline{q}})| \leq V$, which has the form of the inequality `Wall.SSIneq` of [LANA-iut]; there, Step (xi) is proved equivalent to it in lattice models (`xi_iff_ss_model`, file `Iut/WallEquiv/Basic.lean`). In the second model of (Ind2) the comparison in reading (V) is $(s-1)\,|\log(\underline{\underline{q}})| \leq V + \Delta_{\mathrm{V}}(D)$, which has the form of the variant of Corollary 3.12 of [LANA-iut]. There the gap $\Delta(D)$ between the variant and `SSIneq` is proved to be a weighted average of (Ind2)-terms, whose value is left open (file `Iut/WallEquiv/Gap.lean`); [Gap] determines them, and shows that $\Delta(D) > 0$ for every initial Θ-datum and that $\Delta(D)$ is bounded by a quantity that does not involve the size of the q-orders. The proof of Theorem 1.10 of [IUT4] bounds the container further, in terms of the log-different and the log-conductor, to arrive at its final inequality, a height inequality; the program does not carry out that step. The same reduction, with an estimate of the hull in place of V, is the theorem `corThreeTwelve_of_thmThreeElevenFive` in the Lean port of [Richards]. The program decides the outcomes by evaluating the two exact combinations in double precision; every comparison it reports differs from equality by at least 0.87.

4 Windows

Table 1 lists the windows of the program, and Table 2 shows how the outcome of Step (xi) for D_{211} is decided as the global window grows.

window	what is truncated	what growing it shows
p-adic digits K	the unit part of q	lower digits stay fixed and higher digits are added (for D_{211}: $170, 29, 44, 156, \dots$)
global factorization (trial bound, seconds of elliptic-curve factorization, known factors)	the primes dividing the norms	$\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ and the container grow, and the outcome becomes decided (Table 2)
Kummer levels $n \mid p-1$	the Kummer classes of the theta values	the read-out stops changing once the modulus exceeds the value
terms of $\log(1+2\varpi)$	the logarithm series	the valuation of the partial sum stops changing once the term of least valuation is included, at m^* of Section 3.4 (for D_2 at 11: 1, then -94)
depth of (Ind3)	sublattices admitted as possible images	the hull does not change
$\lvert n\rvert \leq W$ in Remark 3.11.1(vii)	the trivial multiradial representation $\{\underline{\underline{q}}^{\lambda^n}\}$	the upper bound of the possible log-volumes tends to 0 (Section 6.1)

Table 1: The windows of the program.

global window	$\lvert\log(\underline{\underline{q}})\rvert$	container $\geq$	outcome in reading (V)
trial division to 10^4	11.377	32.690	undecided (deficit 9.03)
trial division to 10^6, 1 s of factorization	12.184	57.910	holds, with the bound for the unfactored part
with the complete factorization	21.801	236.997	holds

Table 2: Growing the global window for D_{211}. The set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ grows with the window, so $\lvert\log(\underline{\underline{q}})\rvert$ changes as well; each row is a valid choice of $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ in the sense of Definition 3.1.

For an unfactored cofactor C all of whose prime factors exceed the trial bound B, the number Ω of prime factors counted with multiplicity is at most $\log C/\log B$, so $\mathrm{rad}(C) \geq C^{1/\Omega}$. With the formula for e this gives a lower bound for the contribution of the places above C to the container, which the program rounds down to a rational number and reports separately.

5 Results on seven initial Θ-data

Table 3 lists the two sides of Step (xi) for each datum and reading.

datum	p	$-\lvert\log(\underline{\underline{q}})\rvert$	$V \geq$	$-\lvert\log(\underline{\underline{\Theta}})\rvert \geq$		
				(V)	(L)	(L^0)
D_{211}	211	−21.80	237.00	135.26	198.84	217.92
D_{109}	109	−19.11	205.85	116.65	172.40	189.12
D_2	30965771	−17.15	238.03	158.00	199.83	212.38
D_3	30965771	−80.59	825.32	449.24	638.76	695.62
D_{18}	30965771	−43.93	490.90	285.92	412.80	450.87
D_{15}	30965819	−36.86	415.33	243.31	350.82	383.07
$D_{2357}^{(157)}$	2357	−11.84	2062.29	−22413.48	2060.47	2061.38

Table 3: The two sides of Step (xi) on the seven data of [Cat], with $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ as in Section 3.1, in the first model of (Ind2) of Section 3.6, with the complete factorizations of the norms where they were found (for D_3 from [Cat]; for $D_{211}, D_{109}, D_{18}, D_{15}$ by elliptic-curve factorization, the factors being probable primes by the BPSW test). For $D_{2357}^{(157)}$ a cofactor of 1564 digits remains, and V and the lower bounds include the lower bound 261.75 for its contribution (Section 4). The datum $D_{2357}^{(157)}$ is conditional on a report in preparation, as in [Cat].

In readings (L) and (L^0) the computed inequality holds for all seven data, and in reading (V) for all but $D_{2357}^{(157)}$, in each case from the container of the factored places alone. For $D_{2357}^{(157)}$ in reading (V) it is undecided in the window: the deficit is 22401.6, which the places above the 1564-digit cofactor would cover if its radical exceeded about 10^{521}; its prime factors all exceed 10^6 and it is not a prime power. As said in the introduction, none of this bears on Corollary 3.12.

Table 4 lists the gap $\Delta_R(D)$ between the two models of (Ind2) (Section 3.6).

datum	bad places (small, not small)	$\Delta_{\mathrm{V}}(D)$	$\Delta_{\mathrm{L}}(D)$	$\Delta_{\mathrm{L}^0}(D)$
D_{211}	4, 1	25.894–27.253	12.947–13.796	6.473–7.578
D_{109}	6, 0	22.707	11.353	5.677
D_2	3, 2	28.516–30.914	18.251–20.649	12.400–14.798
D_3	14, 5	93.174–102.006	52.581–60.758	33.003–41.788
D_{18}	14, 3	53.621–57.936	28.020–31.821	14.501–18.559
D_{15}	13, 2	45.765–49.165	22.882–26.000	11.441–15.267
$D_{2357}^{(157)}$	3, 6	15.114–28.219	0.029–4.294	0.015–4.507

Table 4: The gap $\Delta_R(D)$ between the shrinkages of the two models of (Ind2), with the data of Table 3. A single value is exact (all bad places small); otherwise the interval comes from the bracket for κ_v at the places that are not small.

In reading (V) the gaps are those of [Gap, §5], whose table bounds them from above with the constants a_v, b_v alone; the bound $\rho_v - \rho'_v$ lowers the upper ends for D_{211}, D_3 and D_{18} (for D_{211} to 27.253 from 27.933). The gaps decrease from (V) to (L) to (L^0): with more positions allowed, the minimum over the positions is more often taken at a good place, where the theta value is 1 and the two models agree. Since the second model has the smaller shrinkage, the comparison holds in it whenever it holds in the first. Over the three global windows of the page (Section 6) and the three readings, the outcomes differ in one case: for D_3 in reading (V) and the intermediate window, the first model leaves the comparison undecided with deficit 0.87, while the second decides that it holds. For $D_{2357}^{(157)}$ in reading (V) the second model lowers the deficit by Δ_{V}, which does not decide it.

Replacing $\mathrm{ord}_w(q)$ by $c \cdot \mathrm{ord}_w(q)$ at every bad place while keeping the container (a hypothetical input, not a datum) multiplies $|\log(\underline{\underline{q}})|$ and, in the first model of (Ind2), Σ_R by c, so the computed inequality holds exactly when $c \le c^* = V/(\Sigma_R - |\log(\underline{\underline{q}})|)$; for D_{211} in reading (V), $c^* = 2.965$. Since V is only a lower bound, the true threshold is at least c^*.

6 The interactive page

The page is a single HTML file that needs nothing but a browser; its text is in Japanese. It has four tabs.

1. *Data*: the comparisons of Step (v) of [IUT4, Thm. 1.10, proof] from [Cat], for each datum and reading.
2. *Running the computation*: the steps of Section 3 with inputs (datum, reading, model of (Ind2), global window, the hypothetical factor c, p-adic digits, Kummer levels) and their outputs, including the number line of Step (xi) and the toys of Section 6.1.
3. *Animation*: the same computation as an animation. Its composition (two circles for the symmetries of the labels, Frobenius-like lines opening into étale-like planes, a triangle of log-shells with the indeterminacies) follows the animation of E. Farcot [Farcot], made from specifications of S. Mochizuki and suggestions of I. Fesenko and licensed under CC BY 4.0. The pictures and motions are new: the numbers, colours and sizes shown, and the positions on the number lines, are taken from the computation on the chosen datum, while the layout and the timing are design choices; the enumeration of tuples (Figure 2) and the final number line do not occur in the original. This adaptation is not endorsed by the authors of the original.
4. *Factory*: the theory drawn as a factory that processes an initial Θ-datum (Figure 1): two halls (the theaters) with the same blueprint (the algorithm) and their own tools (the

rings), a wall (the Θ-link) with a slot for the cable, and an inspection desk (Step (xi)) with the rule (xi-f) in a separate box. Clicking a station opens the computation inside it.

The heavy computations (Tate parameters, logarithm series, Kummer read-outs, factorizations) are exported by the program to a data block in the page. Each time it is opened, the page recomputes the containers, the shrinkages and $|\log(\underline{\underline{q}})|$ as exact combinations of logarithms and checks that they, and the outcomes, agree with the program for all $7 \times 3 \times 3$ combinations of datum, global window and reading, in both models of (Ind2); it enumerates the tuples of places above p for $j \leq 3$ and checks the sums against the closed form (63 capsules); and it displays the program's own checks that the local data reproduce Table 1 of [Cat] and that the log-shell radii agree with the valuations of the series.

6.1 The toys of Remark 3.11.1

Remark 3.11.1(vi)–(viii) of [IUT3] illustrates IPL and SHE by naive constructions, which the program runs in the same frame, at one place w of $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ and with $\lambda = 2$. The naive Θ-link $q \mapsto q^{\lambda}$ within one ring runs, with SHE holding tautologically, but links nothing (no analogue of IPL). With distinct labels ${}^{\dagger}R$ and ${}^{\ddagger}R$ the program cannot compute the comparison and raises the type error of Section 2.1; Remark 3.11.1(vii) says that it is "by no means clear, at least in any a priori sense, how to 'compute' the relationship" between the two arithmetic holomorphic structures. With the forced identification of the two ring structures the output is the trivial multiradial representation $\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}$; in the window $|n| \leq W$ the upper bound of the possible log-volumes is $-\lambda^{-W} c_w r_w \log p_w$, which tends to 0, so the representation says nothing about log-volumes, as Remark 3.11.1(viii) explains.

7 What is computed, given and chosen

Computed. The places, valuations and Tate parameters of the curve; the conditions of Definition 3.1 that are decidable from them; the log-shell radii, with a check of the series; the Kummer read-outs; the containers, possible images, hulls and log-volumes, exactly, in the first model of (Ind2) of Section 3.6, and in the second model exactly at the small places and as an interval elsewhere; the comparison of Step (xi); the gap between the two models; the toys.

Given. That the theta values are $\underline{\underline{q}}^{j^2}$ up to roots of unity, from the theory of the étale theta function as used in [IUT2]; the formula for e of [Cat] at places other than those above p; the log-volume formula for the hull of a tensor packet of [LANA-iut]; the closed form of the hull in the second model of (Ind2), proved on paper in [Gap]; the archimedean container of [IUT4, Thm. 1.10, proof, Step (vii)]; the conditions of Definition 3.1 that the program does not check, such as the image of Galois containing $\mathrm{SL}_2(\mathbb{F}_\ell)$, which are established in [Cat] and the sources cited there.

Chosen. The reading (Section 2.3); the set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$, which leaves out the places above 3 and 5 (Section 3.1); the places above $2, 3, 5, \ell$, whose contributions are set to 0.

Limitations. The anabelian reconstructions are not implemented: the theaters are given as typed copies of fields, not reconstructed from groups, and the Kummer read-out is a round trip. The records of APT and SHE hold by construction. The group of (Ind2) of the source is not modelled; by [LANA-iut], the two models bracket it. In the second model, $\kappa_v(\theta)$ at the places that are not small is only bracketed, since the program does not compute the log-shells as lattices in K_v, nor the unit parts of the theta values at places other than those above p. The model measures the theta values by their valuations in a container shared by the two

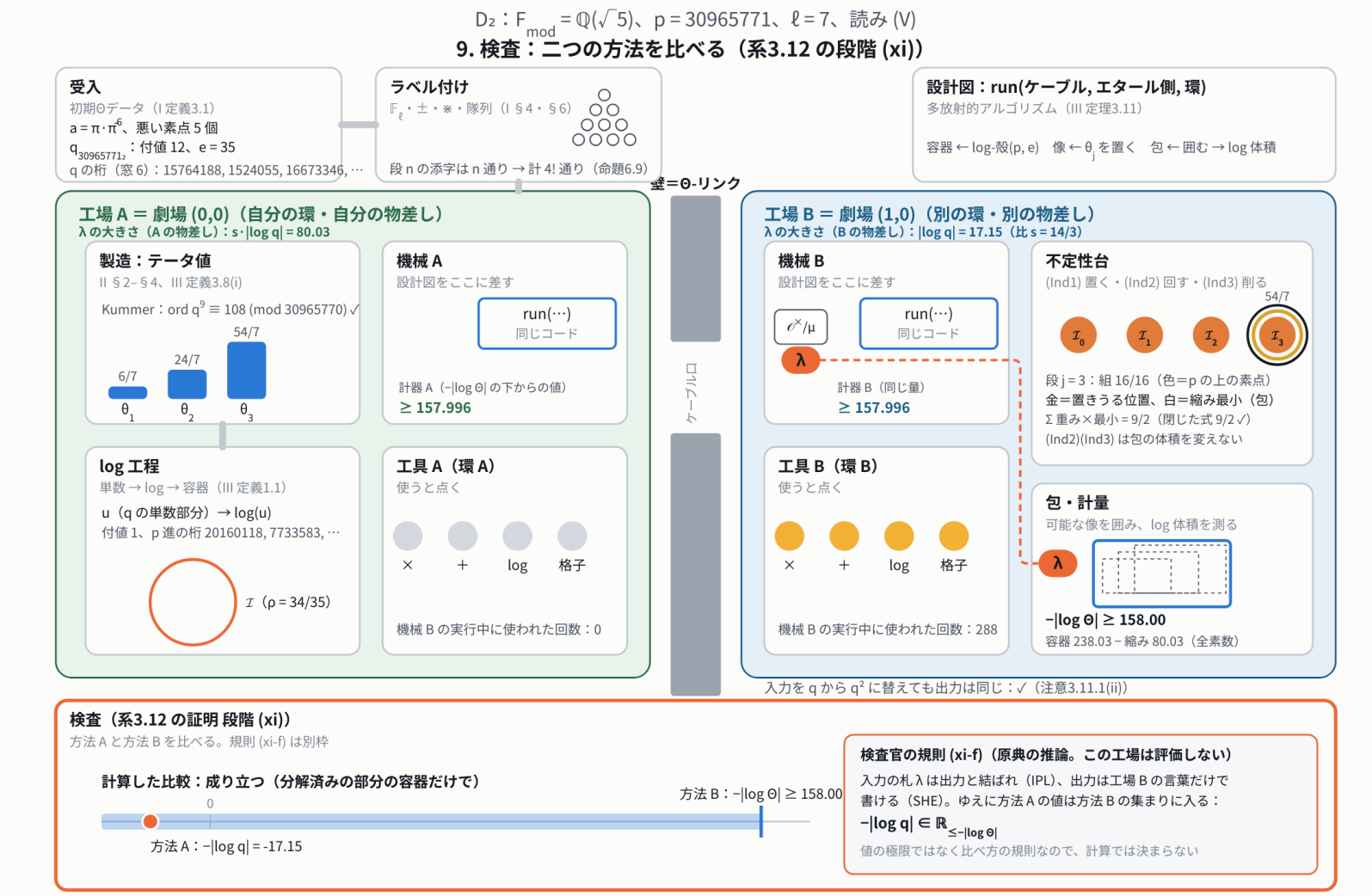


Figure 1: The factory tab for D_2 in reading (V), at the inspection step (labels in Japanese). Left: hall A, the theater $(0,0)$; right: hall B, the theater $(1,0)$; between them the wall (the Θ-link) with the slot through which the cable passes. The same algorithm `run(···)` is installed in both halls and gives the same lower bound 157.996 in both (APT). While machine B ran, the ring operations of hall A were used 0 times and those of hall B 288 times (SHE), and replacing the input q by q^2 did not change the output. The dashed thread ties the object λ that passed the wall to the output (IPL). The indeterminacy table shows the capsule of label 3; the hull box shows $-|\log(\underline{\Theta})| \geq 158.00 = 238.03 - 80.03$. At the bottom, the two numbers of Step (xi) on a line, and, in a separate box, the rule (xi-f), which the page does not evaluate.

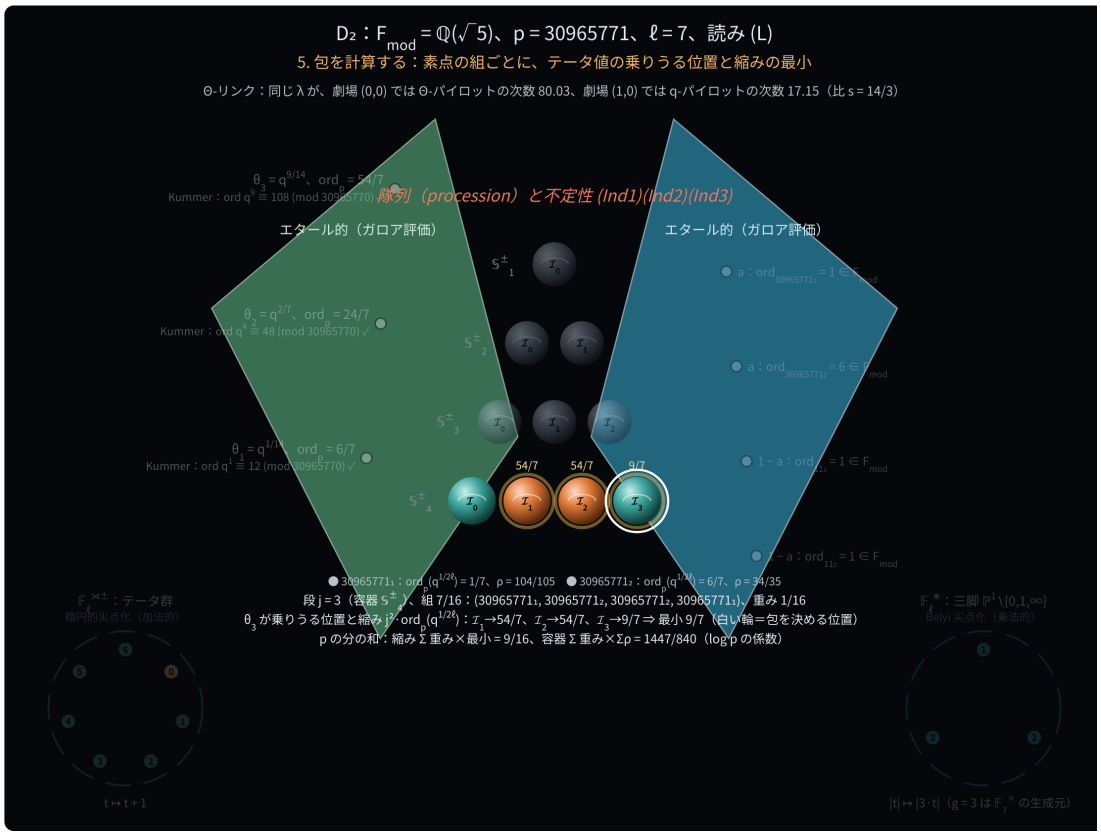


Figure 2: The animation tab for D_2 in reading (L), during the enumeration of the capsule of label 3 (labels in Japanese). The factors $\mathcal{I}_0, \dots, \mathcal{I}_3$ are coloured by the place above p in the current tuple, the seventh of sixteen, of weight $1/16$. Gold rings mark the positions where the theta value may be placed in this reading, with the shrinkage $j^2 \operatorname{ord}_p(\underline{\underline{q}}_{v_t})$ above them (54/7, 54/7, 9/7); the white ring marks the least shrinkage, which determines the hull. The running sums at the bottom are checked against the closed form at the end of the capsule. The composition follows [Farcot].

columns; this is the bookkeeping of the proof of Theorem 1.10 of [IUT4], and the one in which the comparison takes the form of a height inequality. A reader who holds that the output of the algorithm must be measured in another way will not find that view implemented here. The inference (xi-f) is not evaluated.

8 Related work

The toolkit `motizukipdf` of D. Richards [Richards], published in October 2026, turns the objects of IUT I–IV, of [EssLgc] and of related papers into C data types with validators, a registry of 73 processes, a terminal interface, a web application that runs the C code in the browser, and a Lean port. Its coverage is far broader than ours. Its data model lets a Θ-link share the unit groups but never the ring structures (in its Lean port, `wellFormed_not_ring`), which is the first design rule of this note, and records (Ind1) as the $(j+1)!$ permutations of tensor factors. Its initial Θ-data are generated from abc triples via Frey curves over $\mathbb{Q}$, or synthetically; the conditions of Definition 3.1 that can be decided from numbers are checked, and the others are recorded as asserted flags. Its log-volume report for Corollary 3.12 uses the explicit constant of [IUT4, Thm. 1.10] and a leading-term estimate of the hull. With this estimate its process `Fuzzify` widens the Θ-pilot to an interval and reports whether the interval contains $-|\log(\underline{\underline{q}})|$, a numerical form of the comparison of Step (xi) in the terms of [EssLgc, §3.10]. Its arithmetic is in 64-bit integers and floating point; by design, the Tate parameter enters through its valuation and the Kummer isomorphisms as recorded properties (its data model tracks what is identified with what, not the reconstructions). Its Lean port follows the main strategy of [Formal]: it proves IPL (`mltAlgo_IPL`) and APT (`dsc_APT`) for its model, defines SHE as an operation, and derives from the comparison in the common container, left as a hypothesis, the bound $|\log(\underline{\underline{q}})| \le H/(N-1)$ (`corThreeTwelve_of_thmThreeElevenFive`), with H its estimate of the hull term and $N = \ell(\ell+1)/12$: the form of our comparison in reading (V). The present program was written independently; it differs in using verified explicit data whose field of moduli is not $\mathbb{Q}$, in integers of arbitrary size, with the Tate parameters computed p-adically and a Kummer read-out of the theta values (a round trip, Section 3.5); in computing the hull of the union of the possible images exactly, for three readings; in its windows; in running the toys of Remark 3.11.1(vi)–(viii) in the same frame as the algorithm; and in recording IPL, SHE and APT as the execution of one program on numbers.

The analysis of the step from Theorem 3.11 to Corollary 3.12 by the LANA project [LANA] and its Lean repository [LANA-iut] underlies Sections 2.3, 3.6 and 3.7: the log-volume of the hull of a tensor packet of log-shells, the readings of IUT IV and the literal reading together with the minimum versus the average, and the equivalence, in its models, of Step (xi) with an inequality of the form considered by Scholze and Stix [SS]. Its file `Iut/WallEquiv/Gap.lean` reduces the difference between this inequality and its variant of Corollary 3.12 to (Ind2)-terms and leaves their value open; the author's note [Gap] determines them, and Section 3.6 uses its closed form. Mochizuki's report [Formal] describes the formalization of the step from Theorem 3.11 to Corollary 3.12 in Lean as a means of communicating the logical structure of the theory, and names IPL and APT among its central aspects. The permutations of tensor factors in (Ind1) are discussed in [DH1, §4] and [DH, §6.2, footnote 7]. Explicit initial Θ-data over $\mathbb{Q}$ are given in [DH1, §2.10] and [Zhou1, Zhou2], and explicit estimates in [MFHMP]. Expositions of the theory include [Hoshi1, Hoshi2, Yamashita]. A different loss in the hull of a tensor packet, from boxing the integral structure, is studied in [Zeng]; the question of the source of $\lambda = \mathrm{ord}(q^{j^2})$ in Step (v), which the program takes as given (Section 7), is raised in [Takkun]. The data used here and their local invariants are those of the author's catalogue [Cat], which collects data from [First, Second, Note]; its §5 checks the comparison in reading (V) for D_2 with a lower bound for the container, and Table 3 extends this to all seven data.

Use of generative AI

Generative AI tools (large language models) were used extensively in this work, under the author's direction, in writing the program and the web page, in the computations and in the writing. The author takes full responsibility for the contents of this note.

A Reproducing the results

The program is a single Python file (Python 3.10 or later with `sympy`). The commands

```
python3 iut_compute.py --all --reading V     # also L and L0
python3 iut_compute.py --data D2 --toys
python3 iut_compute.py --windows --data D211
python3 iut_compute.py --export-json run_data.json
```

print the steps for all data, the toys, the windows, and the data block of the page. Each run of `--all` takes about fifteen seconds, and the export a little over a minute. Both models of (Ind2) are printed in every run. The recorded outputs are in the supplement, with the page and checksums. The output of the program is in Japanese.

B The code and the source

source	code
IUT I, Definition 3.1; Example 3.2(iv)	`InitialThetaData`, `checks`, `table1_check`
IUT I, §§4, 6; Propositions 6.8, 6.9	`Labels`; the procession counts of the page
Hodge theaters	`Theater` (typed copies), `Elem`, `Lattice`
IUT I, Corollary 3.7; IUT III, Definition 3.8	`theta_link`, `Cable`, `ValuePart`
IUT III, Definition 1.1	`padic_log_unit`, `log_shell_radius`, `LogShell.verify`
Kummer theory	`kummer_recover`
IUT III, Theorem 3.11; Remark 3.11.1	`IUTAlgorithm.run`, `placements`, `run_with_record`
(Ind2): the two models; [Gap, Thms. 3.2, 3.3]; IUT IV, Proposition 1.2(i)	`IUTAlgorithm(ind2=...)`, `kappa_range`, `log_shell_ab`, `log_shell_inner`
IUT III, Corollary 3.12, Step (xi)	`integrate`
IUT III, Remark 3.11.1(vi)–(viii)	`NaiveLink`, `DistinctLabels`, `ForcedIdentification`

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